Supplementary Information for
Exceptional-Point Stability Boundaries from Quantum Dissipation to Cosmological Acceleration

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Contents

Note on revised content. Sections 2 and 7 are revised from the original submission. Section 2.3–2.4 is replaced with a numerical verification showing that the two natural entropy-production proxies for the Gaussian channel model do not produce an interior maximum near $\chi = 1$; the abstract lemmas (Proposition S2.1 and Lemma S2.1) are retained as conditional results. Section 7 is revised accordingly. All other sections are unchanged from the submitted version.

1 Extended Mathematical Derivations

1.1 Lindblad-to-Second-Order Mapping

Here $\omega > 0$ denotes the characteristic frequency magnitude. The dimensionless ratio

$$\chi \equiv \frac{\gamma}{2\omega} \tag{1}$$

plays an analogous role in dynamical stability. Throughout, the critical boundary corresponds to the vanishing of the quadratic discriminant in the associated second-order response.

The GKSL master equation for a harmonic oscillator with amplitude damping

$$\dot{\rho} = -i[H, \rho] + \gamma\left(a\rho a^\dagger - \tfrac{1}{2}\{a^\dagger a, \rho\}\right), \qquad H = \omega a^\dagger a, \tag{2}$$

yields first-moment evolution

$$\dot{x} = -\frac{\gamma}{2}x + \omega p, \qquad \dot{p} = -\omega x - \frac{\gamma}{2}p, \tag{3}$$

where $x = \langle a + a^\dagger \rangle$ and $p = -i\langle a - a^\dagger \rangle$. Differentiating the first equation and substituting the second gives

$$\ddot{x} = -\gamma\dot{x} - \left(\omega^2 + \frac{\gamma^2}{4}\right)x, \tag{4}$$

i.e.,

$$\ddot{x} + \gamma\dot{x} + \Omega^2 x = 0, \qquad \Omega^2 \equiv \omega^2 + \frac{\gamma^2}{4}. \tag{5}$$

This is an exact rewriting of the first-moment dynamics.

For the scalar second-order mode, the critical (double-root) condition is $\gamma^2 - 4\Omega^2 = 0 \Leftrightarrow \gamma = 2\Omega$, at which point $h(t) \propto te^{-\Omega t}$.

Clarification. Under $\Omega^2 = \omega^2 + \gamma^2/4$, imposing $\gamma = 2\Omega$ forces $\omega^2 = 0$. The 2×2 system for (x, p) has eigenvalues $-\gamma/2 \pm i\omega$ and is not defective for $\omega \neq 0$. Exceptional-point behavior in extended Lindblad settings arises under coupled modes, gain/loss imbalance, or reduced effective channels, and is asserted here at the level of the shared quadratic response structure.

1.2 Quadratic Eigenproblem in Open QFT

For the standard damped mode $\ddot{q} + \gamma\dot{q} + \Omega_0^2 q = 0$, the quadratic eigenproblem gives roots

$$\Omega_\pm = -\frac{i\gamma}{2} \pm \sqrt{\Omega_0^2 - \frac{\gamma^2}{4}}. \tag{6}$$

They coalesce when $\Omega_0^2 - \gamma^2/4 = 0 \Leftrightarrow \gamma = 2\Omega_0$, at which point the retarded propagator develops a second-order pole, $G_R(\Omega) \propto (\Omega + i\Omega_0)^{-2}$, and $h(t) \propto te^{-\Omega_0 t}$.

1.3 Cosmological Growth Equation: Full Derivation

The growth equation for matter perturbations:

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G\rho_m\delta = 0. \tag{7}$$

Identifying $\gamma_\delta = 2H$ and $|\omega_\delta| = \sqrt{4\pi G\rho_m}$:

$$\chi_\delta = \frac{H}{\sqrt{4\pi G\rho_m}}. \tag{8}$$

In flat ΛCDM, $\Omega_m = 8\pi G\rho_m/(3H^2)$ gives $4\pi G\rho_m = \frac{3}{2}H^2\Omega_m$, so $\chi_\delta = \sqrt{2/(3\Omega_m)}$. Setting $q = 0$ in $q = \Omega_m/2 - \Omega_\Lambda$ with $\Omega_m + \Omega_\Lambda = 1$ gives $\Omega_m = 2/3$, hence $\chi_\delta = 1$. The identity $\chi_\delta = 1 \Leftrightarrow q = 0$ is established within flat ΛCDM.

2 Information Efficiency: Formal Results and Numerical Scope

Revision note. The original Section 2.3–2.4 claimed that the representative Gaussian channel model (with the kinetic entropy proxy of Eq. 13) exhibits an efficiency maximum near $\chi = 1$ and reported $\chi^* \approx 1.02 \pm 0.05$. Full numerical integration of the exact expressions for both proxy definitions shows this claim is incorrect: neither proxy produces an interior maximum near $\chi = 1$ for any tested bandwidth configuration. The output-power proxy and most kinetic-proxy bandwidths peak at the lower scan boundary (strongly underdamped limit), while the kinetic proxy at $B/\omega_0 = 1$ increases monotonically through the tested scan range. The abstract propositions and lemmas are retained but their scope is revised accordingly.

2.1 Local maximality criterion

Proposition S2.1 (Local maximum criterion)**.** *Let $\eta(\chi) = I(\chi)/\Sigma(\chi)$ with $I, \Sigma \in C^2$ in a neighborhood of χ_0 and $\Sigma(\chi) > 0$. If $\eta'(\chi_0) = 0$ and $\eta''(\chi_0) < 0$, then η admits a strict local maximum at χ_0.*

Proof. Standard second-derivative test. □

The stationarity condition is equivalent to $(\ln I)'(\chi_0) = (\ln \Sigma)'(\chi_0)$. A sufficient condition for strict local maximality is

$$\frac{I''(\chi_0)}{I(\chi_0)} < \frac{\Sigma''(\chi_0)}{\Sigma(\chi_0)}. \tag{9}$$

2.2 Localization under bounded perturbations

Lemma S2.1 (Localization stability)**.** *Let $\eta(\chi)$ be three times continuously differentiable on $[1-\Delta, 1+\Delta]$. Assume $\eta'(1) = 0$, $\eta''(1) = -\kappa < 0$, and $\sup|\eta'''| \le M < \infty$. Then $\chi = 1$ is a strict local maximizer, and every stationary point $\tilde{\chi}$ of the perturbed functional $\tilde{\eta} = \eta + r$ with $\sup|r'| \le \varepsilon$ satisfies $|\tilde{\chi} - 1| \le 2\varepsilon/\kappa$ provided ε is small enough.*

Proof. By Taylor's theorem, $\eta'(\chi) = -\kappa(\chi - 1) + \rho(\chi)$ with $|\rho(\chi)| \le M(\chi - 1)^2/2$. Choosing $\delta \le \kappa/M$ ensures $\eta'(\chi)$ has the sign of $-(\chi-1)$ on $(1-\delta, 1+\delta)\setminus\{1\}$, giving strict local maximality. For the perturbed functional, $\kappa|\tilde{\chi} - 1|/2 \le |r'(\tilde{\chi})| \le \varepsilon$ yields the bound. □

Scope of Lemma S2.1. This result is a conditional localization theorem. It states: *if* a local maximum of an efficiency functional exists at $\chi = 1$ for some channel specification, then small perturbations (finite memory, finite bandwidth, measurement filtering) keep that maximum within an $O(\varepsilon/\kappa)$ neighborhood of $\chi = 1$. The lemma does *not* assert that such a maximum exists for specific proxy definitions. Whether the stationarity conditions hold depends on the channel parameterization.

2.3 Gaussian channel setup

Consider the stable second-order channel

$$\ddot{x} + 2\chi\omega\dot{x} + \omega^2 x = u(t), \tag{10}$$

with transfer function $H(i\Omega;\chi) = (\omega^2 - \Omega^2 + 2i\chi\omega\Omega)^{-1}$. The input u has flat PSD S_0 on $|\Omega| \le B$; the output is corrupted by white noise with PSD N_0.

Information throughput:

$$I(\chi) = \frac{1}{4\pi}\int_{-B}^{B} \ln\left(1 + \frac{S_0}{N_0}|H(i\Omega;\chi)|^2\right) d\Omega. \tag{11}$$

Proxy definitions:

$$\Sigma_{\text{pow}}(\chi) = \kappa(2\chi\omega)\int_{-B}^{B} |H(i\Omega;\chi)|^2 \, d\Omega, \tag{12}$$

$$\Sigma_{\text{kin}}(\chi) = \kappa(2\chi\omega)\frac{S_0}{2\pi}\int_{-B}^{B} \Omega^2 |H(i\Omega;\chi)|^2 \, d\Omega. \tag{13}$$

The proxy Σ_{kin} is proportional to $\chi \cdot \mathbb{E}[\dot{x}^2]$ under stationary driving.

2.4 Numerical verification: both proxies lack interior maximum near $\chi = 1$

Full numerical integration of $\eta_{\text{pow}}(\chi) = I(\chi)/\Sigma_{\text{pow}}$ and $\eta_{\text{kin}}(\chi) = I(\chi)/\Sigma_{\text{kin}}$ was performed across a dense χ grid $[0.03, 3.0]$ for bandwidth ratios $B/\omega \in \{0.5, 1, 2, 5, 10\}$ with $S_0/N_0 = 10$.

Table 1: Location of $\eta(\chi)$ maximum for both proxy definitions across bandwidth ratios. "Boundary" indicates the maximum is at the lower scan boundary ($\chi \approx 0.03$), not an interior maximum.

Proxy	B/ω_0	χ^*	Interior maximum near $\chi = 1$?
Σ_{pow}	0.5, 1, 2, 5, 10	≈ 0.03 (boundary)	No
Σ_{kin}	0.5, 2, 5, 10	≈ 0.03 (boundary)	No
Σ_{kin}	1.0	monotone increasing through scan; $\chi > 3$	No

In all tested cases, neither proxy produces an interior maximum near $\chi = 1$. For the output-power proxy and the kinetic proxy at $B/\omega_0 \in \{0.5, 2, 5, 10\}$, the maximum occurs at the lower scan boundary (strongly underdamped limit). For the kinetic proxy at $B/\omega_0 = 1$, $\eta_{\text{kin}}(\chi)$ increases monotonically throughout the scan range $[0.03, 3.0]$ and continues rising beyond $\chi = 3$; no interior maximum is found within the tested domain. Thus neither proxy supports the original claim of an efficiency optimum near $\chi = 1$.

The approximate formula used in the original Section 7.1,

$$\eta(\chi) \approx \frac{C\log_2(1 + D/\chi)}{\Sigma_0(1 + \alpha\chi^2)}, \tag{14}$$

uses an underdamped-regime approximation for I and an overdamped-regime proxy for Σ. Numerical optimization of Eq. (14) confirms a maximum near $\chi \approx 0.05$ for all tested (D, α) combinations, not near $\chi = 1$. The original claim $\chi^* \approx 1.02 \pm 0.05$ is retracted for these proxy definitions.

2.5 Summary

Lemma S2.1 (localization stability) is a valid conditional theorem. Given that neither tested proxy produces a local maximum at $\chi = 1$, the lemma does not apply to these specific channel models. The main text accordingly removes the claim that $\chi = 1$ is an efficiency-maximizing operating point for the Gaussian channel models analyzed here. The discriminant boundary at $\chi = 1$ remains the exact stability separator for any system admitting the second-order response form, independent of any proxy choice.

3 Conventions for (ω, Γ) in Figure 3 and Table S1

For elementary particles, ω is identified with rest mass in natural units ($\hbar = c = 1$) and Γ with the measured total decay width. For effectively stable particles, Γ is treated as an experimental upper bound or an interaction-limited width under the stated convention.

Specific conventions:

- Massive particles: $\omega \equiv m$; Γ = total decay width (PDG).
- Stable particles: Γ represents the experimental upper bound.
- Vacuum substrates: ω is the characteristic energy scale and $\Gamma = 2\omega$ is imposed by the critical-damping consistency condition $\chi = 1$. **These are defined anchors, not measured decay widths.** The dark energy entry uses the Hubble energy scale as a frequency proxy; dark energy has no measured decay width.
- Confined quarks: Γ represents the hadronization scale $\sim \Lambda_{\text{QCD}}$.

4 QCD Substrate Damping: Order-of-Magnitude Estimates and Falsification Protocol

Scope. The estimates below are order-of-magnitude consistency checks intended to motivate a concrete lattice observable. The term "exceptional point" follows the convention of non-Hermitian physics [11, 12] and is unrelated to "exceptional configurations" in lattice QCD.

4.1 Thermal Baseline from Hard-Thermal-Loop Theory

In the deconfined phase just above $T_c \approx$ 150–170 MeV, HTL effective theory gives the gluon plasmon dispersion $\omega^2 = k^2 + m_D^2 - i\omega\gamma_{\text{HTL}}$ where $m_D^2 = g^2T^2(N_c/3 + N_f/6)$ and $\gamma_{\text{HTL}} = (g^2T/2\pi)[\ln(2T/\omega) + C]$ with $C \sim O(1)$. For $\alpha_s \approx$ 0.3–0.5 at $T \sim \Lambda_{\text{QCD}}$: $\gamma_{\text{HTL}} \sim$ 60–100 MeV, giving a reference scale $\chi_{\text{ref}} = \gamma_{\text{HTL}}/(2\Lambda_{\text{QCD}}) \sim$ 0.15–0.25.

4.2 Effective Damping Scale Near the QCD Transition

For independent incoherent channels, $\max_i(\gamma_i) \leq \Gamma_{\text{QCD}} \leq \sum_i \gamma_i$. Representative magnitudes: $\gamma_{\text{HTL}} \sim$ 60–100 MeV, $\gamma_{\text{inst}} \sim$ 50–80 MeV, $\gamma_{\text{chiral}} \sim$ 10–60 MeV. These are heuristic and non-additive; they illustrate that $\chi \approx 1$ at T_c is not excluded by known scales. Current estimates yield $\Gamma_{\text{QCD}} \sim$ 150–300 MeV, somewhat below the near-critical window $0.8 \lesssim \chi_{\text{QCD}} \lesssim 1.2$ (which requires $\Gamma_{\text{QCD}} \sim$ 320–480 MeV for $\Lambda_{\text{QCD}} \sim$ 200 MeV). This constitutes a quantitative test rather than an assumed input.

4.2.1 Note on Spinodal-like Amplification

At zero baryon chemical potential, lattice QCD indicates a crossover rather than a first-order transition. Strict spinodal decomposition does not generically occur in equilibrium QCD at $\mu \approx 0$. Rapid nonequilibrium cooling or first-order regions of the phase diagram may exhibit spinodal-like amplification; the present framework does not assume spinodal behavior in equilibrium QCD.

4.3 Lattice-Testable Scalar-Channel Protocol

At finite temperature, lattice correlators

$$G(\tau, T) = \int_0^\infty \frac{d\omega}{2\pi} \rho(\omega, T) \frac{\cosh[\omega(\tau - \beta/2)]}{\sinh(\omega\beta/2)} \tag{15}$$

yield the spectral function $\rho(\omega, T)$ via MEM, Bayesian, or Backus-Gilbert reconstruction. The stability index is

$$\chi_{\text{lattice}}(T) = \frac{\Gamma(T)}{2m_0(T)}, \tag{16}$$

where $m_0(T)$ and $\Gamma(T)$ are the dominant spectral peak location and width.

Falsification criterion: The inheritance hypothesis is disfavored if $\chi_{\text{lattice}}(T) < 0.5$ or > 2 throughout the crossover region across all reconstruction methods. Identification of a robust band $0.8 \lesssim \chi_{\text{lattice}}(T) \lesssim 1.2$ near T_c would support the near-critical substrate hypothesis.

5 Illustrative Weak-Mixing Toy Model

The electron mode arises from weak coupling to a near-critical QCD condensate in this dimensional toy model. Diagonalization yields

$$m_e \approx \frac{V^2}{2\Lambda_{\text{QCD}}} \equiv \epsilon_e \Lambda_{\text{QCD}}. \tag{17}$$

Using $\Lambda_{\text{QCD}} \approx 200$ MeV and $m_e = 0.511$ MeV gives $\epsilon_e \approx 2.6 \times 10^{-3}$, and the Yukawa coupling $y_e = \epsilon_e \sqrt{2}\Lambda_{\text{QCD}}/v \approx 2.9 \times 10^{-6}$ reproduces the Standard Model order of magnitude. This is a dimensional illustration of scale separation from weak mixing, not a derivation of the electron mass.

6 Renormalization Group Stability: Multi-Loop Analysis

6.1 One-Loop Calculation

For weakly coupled $\lambda\phi^4$ theory with dissipation:

$$\frac{d\chi}{d\ell} = \chi \frac{\lambda}{32\pi^2}. \tag{18}$$

For $\lambda = 0.1$ over three decades ($\Delta\ell = \ln 10^3 = 6.9$): $\Delta\chi/\chi \approx 2.19 \times 10^{-3}$, confirming $|\Delta\chi| < 0.3\%$ at one loop.

Extension to strongly coupled regimes, including the QCD critical region, requires dedicated functional RG or lattice methods; the perturbative result applies only in weakly coupled models.

6.2 Two-Loop Corrections

The two-loop contribution $d\chi/d\ell|_{2-\text{loop}} = \chi\lambda^2(c_\gamma - c_\omega)/(16\pi^2)^2$ gives $|\Delta\chi|_{2-\text{loop}} \sim 3 \times 10^{-5}\chi_0$ over three decades for $\lambda = 0.1$, entirely negligible.

6.3 Non-Perturbative Stability

In strongly coupled regimes ($\lambda \gtrsim 1$), perturbative RG breaks down. Non-perturbative behavior requires dedicated lattice or functional RG analysis.

7 Information Efficiency: Revised Extended Analysis

Revision note. The original Sections 7.1–7.3 claimed that the Gaussian channel model produces $\chi^* \approx 1.02$ and that non-Gaussian and non-Markovian extensions shift this maximum by $O(10\%)$. Since full numerical integration does not produce an interior maximum near $\chi = 1$ for either tested proxy definition, these claims are revised.

7.1 Gaussian channel: retraction of $\chi^* \approx 1.02$

The mutual information for the linear channel

$$I(\chi) = \int_0^B \log_2\left(1 + \frac{|H(\omega;\chi)|^2 S_0}{N_0}\right) d\omega \tag{19}$$

has $|H|^2$ sharply peaked near $\omega_r = \omega_0\sqrt{1-\chi^2/2}$ for $\chi \ll 1$ (peak value $\approx 1/(2\chi\omega_0^2)$), broad for $\chi = 1$, and monotone for $\chi > 1$.

For the tested Gaussian-channel proxies, no interior maximum near $\chi = 1$ is found. The output-power proxy and the kinetic proxy for $B/\omega_0 \in \{0.5, 2, 5, 10\}$ peak at the lower scan boundary (strongly underdamped limit), while the kinetic proxy at $B/\omega_0 = 1$ increases through the tested scan range with no interior turning point. Thus the original claim of a robust Gaussian-channel efficiency maximum near $\chi = 1$ is retracted for these proxy definitions.

7.2 Implication for non-Gaussian and non-Markovian extensions

The original claims that Lévy noise shifts χ^* by $\sim$ 8–15% and that non-Markovian effects keep χ^* within $[0.85, 1.15]$ were predicated on the existence of a maximum near $\chi = 1$ in the Gaussian baseline. Since that existence is not established for the proxies tested, these robustness claims no longer apply.

Lemma S2.1 (localization stability) retains its value as a conditional result: if any channel specification is found to produce $\eta'(1) = 0$ and $\eta''(1) < 0$, then small perturbations (including non-Gaussian noise and non-Markovian correlations) keep the maximum within an $O(\varepsilon/\kappa)$ neighborhood of $\chi = 1$.

8 Neutrino Sector: MSW Resonances and Collective Effects

8.1 Matter-Induced Dephasing vs. Primordial Hierarchy

The critical damping mechanism addresses mass generation, not propagation. The primordial constraint $\chi_k^{(\mathrm{prim})} \propto \Gamma_{\mathrm{sub}}/m_k^2 \approx 1$ during the formation epoch established the mass ordering $m_1 < m_2 < m_3$. Today, $\Gamma_{\mathrm{sub}} \to 0$ so $\chi_k \to 0$, ensuring coherent oscillations.

8.2 MSW Effect: Orthogonal to Critical Damping

The MSW mechanism modifies propagation eigenstates via coherent forward scattering; the critical damping mechanism sets mass eigenvalues via substrate inheritance. These are orthogonal. MSW operates on $\sim 10^{-23}$ eV scale corrections; the critical damping mechanism operates on the $\sim 10^{-2}$ eV mass scale.

8.3 Collective Neutrino Oscillations

Collective modes exhibit instabilities when $\mu > \omega_{\mathrm{vac}} = \Delta m^2/(2E)$. The mass-ordered hierarchy (set primordially) determines Δm_{ij}^2 and hence the instability threshold. All mass eigenstates satisfy $\chi_k \ll 1$ in any terrestrial or astrophysical medium.

9 Experimental Protocols and Statistical Framework

9.1 Circuit QED: Detailed Protocol

Scope note. The following protocol is illustrative for an engineered dissipative mode or synthetic non-Hermitian two-mode platform. The standard weak-Purcell dispersive transmon formula is not assumed to remain valid at $\chi \sim 1$; at such large effective damping the formula breaks down and the effective damping rate must be calibrated from the measured response rather than inferred analytically.

Setup: Transmon qubit ($\omega_q/2\pi = 5$ GHz) coupled to 3D cavity ($\omega_c/2\pi = 7$ GHz, $\kappa/2\pi = 1$ MHz) with tunable coupling $g/2\pi = 100$–300 MHz.

Procedure: (1) Initialize qubit in $|1\rangle$ via π-pulse. (2) Apply detuning pulse to set $\Delta = \omega_c - \omega_q$. (3) In the weak-dispersive limit, the Purcell decay rate scales as $\gamma_P = \kappa g^2/\Delta^2$; near $\chi \sim 1$, however, the effective damping rate should be calibrated experimentally from the measured response. (4) Effective $\chi = \gamma_P/(2\omega_q)$ tuned by varying g or Δ. (5) Measure $P_1(t)$ every $\delta t = 10$ ns for duration $T = 10\,\mu$s. (6) Fit $P_1(t) = Ae^{-\gamma t/2}\cos(\omega_a t + \phi)$ for $\chi < 1$; $P_1(t) = Bte^{-\omega t}$ for $\chi = 1$. (7) Extract $\omega_a = \omega_q\sqrt{1-\chi^2}$ and γ. (8) Repeat for $\chi \in \{0.5, 0.7, 0.85, 0.95, 1.0, 1.05, 1.15, 1.3\}$.

9.2 Trapped Ions: Protocol

Setup: Single $^{40}Ca^+$ in linear Paul trap. Axial trap frequency $\omega_z/2\pi = 1$ MHz.

Procedure: (1) Laser cool to ground state ($\bar{n} < 0.1$). (2) Apply displacement pulse to coherently displace motional state. (3) Tune cooling laser intensity to set $\gamma = \Gamma_{\text{cool}}$. (4) Monitor amplitude via sideband fluorescence spectroscopy. (5) Vary Γ_{cool} to scan $\chi \in [0.5, 1.5]$. At $\chi = 1$, the motional sideband at ω_z should collapse.

9.3 Statistical Framework

Hypothesis testing: Null H_0: generic damped oscillator. Alternative H_1: EP2 transition at $\chi = 1$. Test statistic $T = |\omega_a(\chi = 1)|/\sigma_{\omega_a}$. Under H_1, $\omega_a \to 0$ and $T \to 0$. Under H_0, $T > 3$ (reject at 3σ).

Bayes factor: $B_{10} = p(D|M_1)/p(D|M_0)$; $B_{10} > 100$ provides decisive evidence for M_1.

10 Finite-Memory and Non-Markovian Extensions

10.1 Exponential Memory Kernel

For $K(t) = \gamma_0 e^{-t/\tau_m}$, the effective damping $\gamma_{\text{eff}}(\omega) = \gamma_0/(1 + \omega^2\tau_m^2)$ gives $\chi_{\text{eff}} = \gamma_0/[2\omega_0(1 + \omega_0^2\tau_m^2)]$. At $\omega_0\tau_m = 1$: $\chi_{\text{eff}} = \chi_{\text{Markov}}/2$. Finite memory broadens the practical transition window without removing the discriminant boundary in Markovian effective reductions.

10.2 Power-Law Memory: Fractional Dissipation

For $K(t) \propto t^{-\alpha}$ with $0 < \alpha < 1$, the effective damping $\gamma_{\text{eff}}(\omega) = \gamma_\alpha\omega^\alpha$ gives $\chi(\omega) = \gamma_\alpha\omega^{\alpha-1}/2$. The critical boundary occurs at $\omega^* = (2/\gamma_\alpha)^{1/(\alpha-1)}$. Non-Markovian effects shift the location of the $\chi = 1$ boundary in frequency space but preserve its existence as the qualitative regime separator.

11 Cross-Scale Validation and Logarithmic Compression

11.1 QCD Sector: σ-Meson

PDG values [6]: mass $m_\sigma = 400$–550 MeV (central: 475 MeV); width $\Gamma_\sigma = 400$–700 MeV (central: 550 MeV). Central stability ratio: $\chi_\sigma = 550/(2 \times 475) \approx 0.58$. Full PDG range: $\chi_\sigma \in [0.36, 0.88]$. The value $\chi = 1$ is within the uncertainty range but above the central value; this entry is near-unity within uncertainties, not at unity.

11.2 Atomic Nuclei: Giant Resonances

For ^{208}Pb: $E_{\text{GDR}} \approx 13.5$ MeV, $\Gamma_{\text{GDR}} \approx 4.0$ MeV. $\chi_{\text{GDR}} = 4.0/(2 \times 13.5) \approx 0.15$. Safely underdamped.

11.3 Neutrinos: Mass Eigenstates

Charged-current rates are suppressed far below $\omega_k = m_k^2/(2E)$. All mass eigenstates satisfy $\chi_k \ll 1$, consistent with coherent oscillations over astronomical baselines.

11.4 One-Sample KS Test: Statistical Analysis

A one-sample Kolmogorov-Smirnov test comparing $\log_{10}(\chi)$ for the empirical subset ($n = 23$, vacuum anchors excluded, photon and gluon excluded) against the analytical uniform distribution over the observed range yields $D = 0.540$, $p = 8.39 \times 10^{-7}$ (verified stable across Monte Carlo null samples of 10^5–10^7 and the analytical one-sample result). Bootstrap confidence intervals (10^4 resamples) give median $\log_{10}\chi = -16.9$ (95% CI: $[-33.3, -12.2]$) for stable matter and -1.0 (95% CI: $[-1.9, 1.0]$) for short-lived resonances ($\chi > 0.01$). The non-uniformity reflects survivorship at $\chi \ll 1$ and a secondary resonance cluster near $\chi \sim 0.1$–1; evidence for the resonance sub-cluster alone is suggestive rather than conclusive.

12 Supplementary Table S1: Particle Data Compilation

Table 2 compiles the complete dataset plotted in Figure 3. Vacuum-scale entries (QCD Scale, Electroweak Scale, GUT Scale, Planck Scale, Dark Energy) are defined anchors satisfying $\chi = 1$ by construction and are excluded from statistical estimates. The dark energy entry uses the Hubble energy scale as a frequency proxy; dark energy has no measured decay width, and $\Gamma = 2\omega_{\mathrm{DE}}$ is imposed by convention.

13 Future Directions

QCD damping: Current HTL-based estimates yield $\Gamma_{\mathrm{QCD}} \sim 150$–300 MeV; the near-critical window requires ~ 320–480 MeV. Lattice falsification protocol is explicit (Section 4.3).

RG stability: Two-loop analysis confirms slow perturbative running. Non-perturbative behavior requires dedicated lattice or functional RG investigation.

Information efficiency: The efficiency maximum near $\chi = 1$ is retracted for the two Gaussian proxy definitions tested. Lemma S2.1 (localization stability) remains valid as a conditional theorem.

Neutrinos: MSW and collective effects are orthogonal to the primordial mass-setting mechanism. All eigenstates satisfy $\chi_k \ll 1$.

Experimental protocols: Circuit QED, trapped ion, and optomechanical procedures are specified in Sections 9.1–9.2 with statistical frameworks in Section 9.3.

Cross-scale validation: One-sample KS test confirms non-uniform χ-space structure ($D = 0.540$, $p = 8.39 \times 10^{-7}$); the sub-cluster evidence near $\chi \sim 1$ is suggestive rather than conclusive.

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Table 2: Supplementary Table S1. $\chi = \Gamma/(2\omega)$. Vacuum-scale entries have $\chi = 1$ by construction and are excluded from statistical tests. All other entries from PDG 2022 [6] and Planck 2020 [9].

System	Mass/Energy (eV)	Width/Damping (eV)	χ	Category
Dark Energy	2.4×10^{-33}	2.4×10^{-33}	1.0 (constr.)	Vacuum
Neutrino (ν_3)	5.0×10^{-2}	—	$\ll 1$	Lepton
Neutrino (ν_2)	8.6×10^{-3}	—	$\ll 1$	Lepton
Electron	5.11×10^{5}	$< 10^{-51}$	$< 10^{-57}$	Lepton
Muon	1.06×10^{8}	3.0×10^{-10}	1.4×10^{-18}	Lepton
Tau	1.78×10^{9}	2.3×10^{-3}	6.5×10^{-13}	Lepton
Up quark	2.2×10^{6}	$\sim 10^{8}$	~ 25	Quark
Down quark	4.7×10^{6}	$\sim 10^{8}$	~ 11	Quark
Strange quark	9.5×10^{7}	$\sim 10^{8}$	~ 0.5	Quark
Charm quark	1.28×10^{9}	$\sim 10^{8}$	~ 0.04	Quark
Bottom quark	4.18×10^{9}	$\sim 10^{8}$	~ 0.01	Quark
Top quark	1.73×10^{11}	1.42×10^{9}	4.1×10^{-3}	Quark
Proton	9.38×10^{8}	$< 10^{-24}$	$< 5 \times 10^{-34}$	Baryon
Photon	0	0	—	Boson
Gluon	0	0	—	Boson
W boson	8.04×10^{10}	2.09×10^{9}	1.3×10^{-2}	Boson
Z boson	9.12×10^{10}	2.50×10^{9}	1.4×10^{-2}	Boson
Higgs boson	1.25×10^{11}	4.07×10^{6}	1.6×10^{-5}	Boson
Pion (π^0)	1.35×10^{8}	7.81	2.9×10^{-8}	Meson
Pion ($\pi^\pm$)	1.40×10^{8}	2.53×10^{-8}	9.0×10^{-17}	Meson
Kaon (K^0)	4.98×10^{8}	7.35×10^{-6}	7.4×10^{-15}	Meson
η meson	5.48×10^{8}	1.31×10^{-3}	1.2×10^{-12}	Meson
ρ meson	7.75×10^{8}	1.49×10^{8}	9.6×10^{-2}	Meson
ω meson	7.83×10^{8}	8.49×10^{6}	5.4×10^{-3}	Meson
ϕ meson	1.02×10^{9}	4.25×10^{6}	2.1×10^{-3}	Meson
σ ($f_0(500)$)	4.75×10^{8}	5.50×10^{8}	0.58	Meson
Neutron	9.40×10^{8}	7.43×10^{-19}	4.0×10^{-28}	Baryon
$\Delta(1232)$	1.23×10^{9}	1.17×10^{8}	4.8×10^{-2}	Baryon
QCD Scale	2.00×10^{8}	4.00×10^{8}	1.0 (constr.)	Vacuum
EW Scale	2.46×10^{11}	4.92×10^{11}	1.0 (constr.)	Vacuum
GUT Scale	$\sim 10^{25}$	$\sim 2 \times 10^{25}$	~1.0 (constr.)	Vacuum
Planck Scale	1.22×10^{28}	$\sim 2.4 \times 10^{28}$	~1.0 (constr.)	Vacuum

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